

HOME TASK

(A)

Given a table of candidates and their skills, you're tasked with finding the candidates best suited for an open Data Science job. You want to find candidates who are proficient in Python, Tableau, and PostgreSQL.

Write a query to list the candidates who possess all of the required skills for the job. Sort the output by candidate ID in ascending order.

CODE:

```
SELECT candidate_id
FROM candidates
WHERE skill IN ('Python', 'Tableau', 'PostgreSQL')
GROUP BY candidate_id
HAVING COUNT(skill) = 3
ORDER BY candidate_id ASC;
```

(B) Assume you're given two tables containing data about Facebook Pages and their respective likes (as in "Like a Facebook Page").

Write a query to return the IDs of the Facebook pages that have zero likes. The output should be sorted in ascending order based on the page IDs.

CODE:

```
SELECT pages.page_id
FROM pages
LEFT JOIN page_likes
  ON pages.page_id = page_likes.page_id
WHERE page_likes.page_id IS NULL
ORDER BY pages.page_id ASC;
```

(C)

Tesla is investigating production bottlenecks and they need your help to extract the relevant data. Write a query to determine which parts have begun the assembly process but are not yet finished.

Assumptions:

- **parts_assembly** table contains all parts currently in production, each at varying stages of the assembly process.
- An unfinished part is one that lacks a **finish_date**.

This question is straightforward, so let's approach it with simplicity in both thinking and solution.

CODE:

```
SELECT part, assembly_step  
FROM parts_assembly  
WHERE finish_date IS NULL;
```